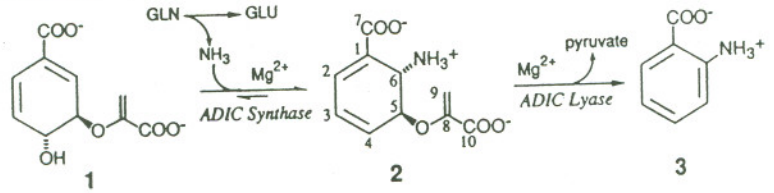
ORGANIC CUMULATIVE EXAM

April 3, 1993

Morollo *et. al.* have recently reported on the isolation and structure determination of 2-amino-2-deoxyisochorismate (**2**), an intermediate in the biosynthesis of anthraniliate (**3**). *J. Am. Chem. Soc.*, **1993**, *115*, 816-817.

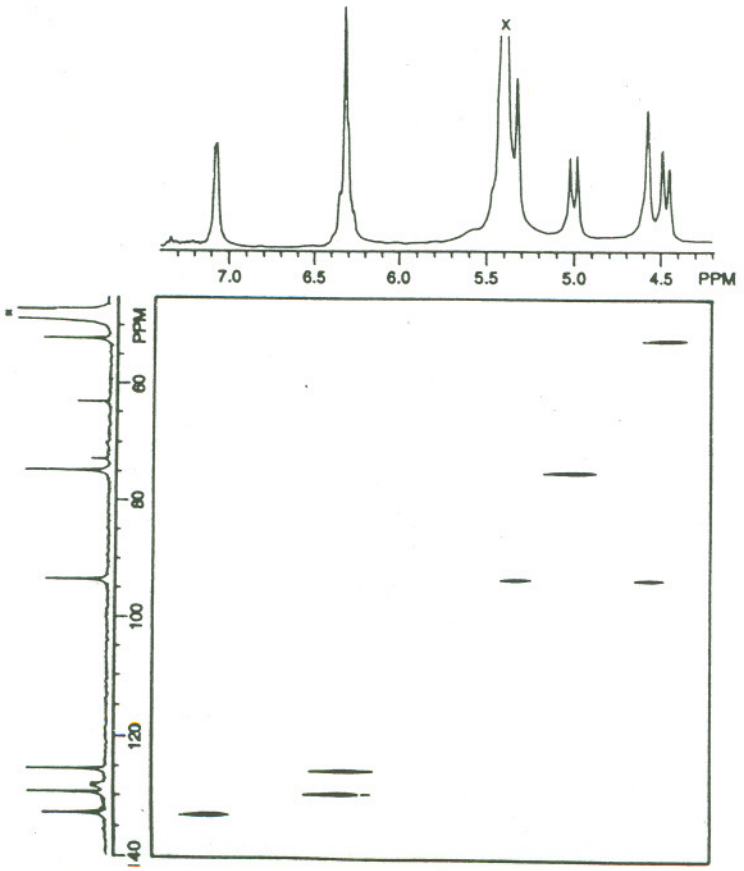
Scheme I



1) Using electron-pushing, show how **1** is converted to **2** by the enzyme ADIC synthase. Assume the enzyme has appropriate acids and bases to carry out any proton transfers needed.

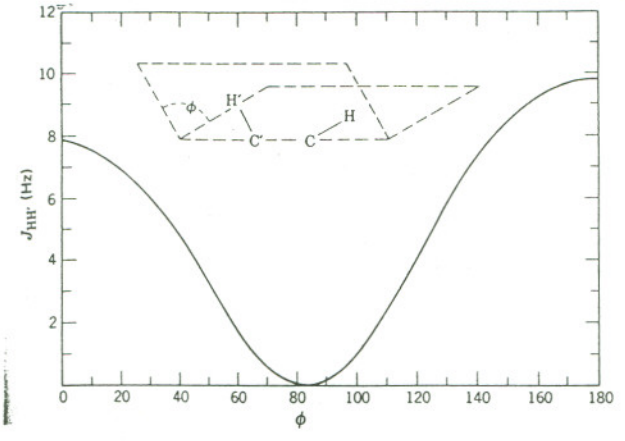
2) Using electron-pushing, show how **2** is converted to **3** by the enzyme ADIC lyase. Assume the enzyme has appropriate acids and bases to carry out any proton transfers needed.

3) Shown below is the $^{1}H,^{13}C$-COSY 2D NMR of **2** at 300 MHz for ^{1}H. Discuss, *in detail*, how the chemical shifts, splitting patterns, and COSY cross peaks can be used to assign the structure to **2**.

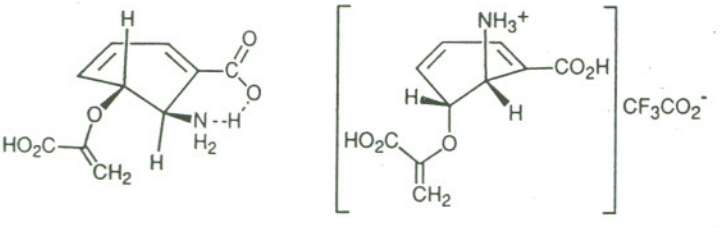


Organic Cumulative Exam (con't)

4) The H5 and H6 coupling constant is 12.3 Hz. Using the accompanying Karplus diagram, explain how the conformational structure of **2** can be assigned as diequatorial or diaxial.



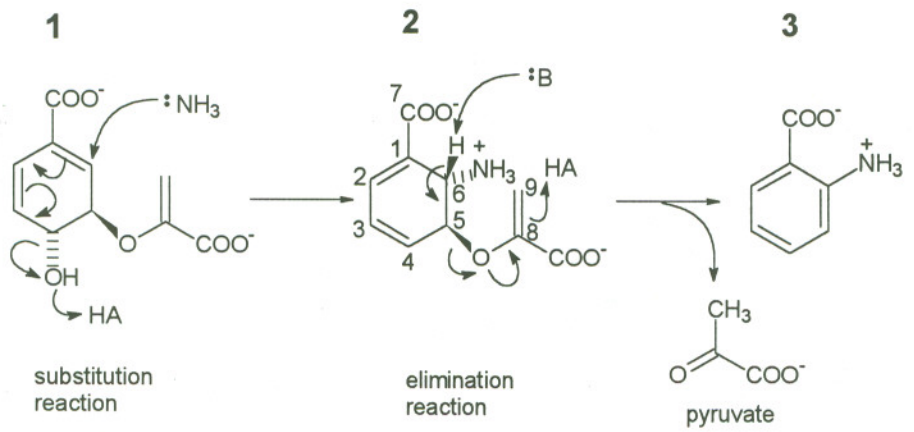
Scheme II



2 (diequatorial) **2•TFA** (diaxial)

Organic Cume
April 3, 1993
Answer Key

Morollo *et. al.* have recently reported on the isolation and structure determination of 2-amino-2-deoxyisochorismate (**2**), an intermediate in the biosynthesis of anthranilate (**3**). *J. Am. Chem. Soc.*, **1993**, *115*, 816-817. Using electron-pushing, show how 1) **1** is converted to **2** by the enzyme ADIC synthase and 2) how **2** is converted to **3** by the enzyme ADIC lyase. Assume the enzyme has appropriate acids and bases to carry out any proton transfers needed.



3) The doublets at δ = 4.45 and 5.0 ppm are assigned to H6 and H5, respectively (O is more electronegative than N). These give ^{1}H, ^{13}C COSY crosspeaks for C5 at 70 ppm and C6 at 52 ppm. The vinyl H9 protons occur at 4.6 and 5.4 ppm with a <u>single</u> COSY cross peak at C9 = 94 ppm in the vinyl ether region of the ^{13}C spectrum. The remaining H2,H3, and H4/C2,C3, and C4 peaks are difficult to assign individually but occur downfield at 6.4-7.1 ppm in the ^{1}H spectrum and 125-135 ppm in the ^{13}C spectrum.

4)The H5 and H6 coupling constant is 12.3 Hz. From the generalized Karplus diagram this corresonds to $\approx 180^{\circ}$bond angle for the two protons at H5 and H6. The conformational structure of **2** can be assigned as **diequatorial** (with the two protons in a diaxial relationship).